

Yu Xiang Luo

☎ +886 908-780-503

✉ linus13514@gmail.com

🌐 [LinkedIn](#)

🐙 [Github](#)

🌐 [Website](#)

Education

National Taiwan University

Sep 2025 – Jun 2027

Master of Science in Computer Science, GPA: 4.22/4.3

Taipei, Taiwan

- Member of Communication & Multimedia Lab (CMLAB), AIMM Group.
- Specializing in Computer Vision, Efficient MLLM inference.

National Taiwan University

Sep 2021 – Jun 2025

Bachelor of Science in Computer Science, GPA: 3.72/4.3

Taipei, Taiwan

Experience

Synology Inc.

Jul 2026 – Aug 2026

Product Developer Intern

New Taipei, Taiwan

- Replaced a caption-then-embed image search pipeline with an 8-bit SigLIP 2 dual encoder on MLX.
- Cut per-image index latency from 583ms to 21ms and peak memory from 4.6GB to 0.9GB on Apple Silicon.
- Raised image retrieval Recall@5 from 48% to 72% on the XM3600 English benchmark.

Yuanta Securities

Sep 2025 – Jan 2026

Quantitative Trading Research Intern

Taipei, Taiwan

- Modeled the IX0208 index to forecast 00940 ETF constituent changes with 80-90% reconstruction accuracy.
- Optimized a competitive bid loan model by stratifying market data and applying tailored strategies to each tier.
- Developed a Robotic Process Automation (RPA) app with GUI workflow builder for traders to scrape data.

Taiwan AI Labs

Apr 2024 – Mar 2025

Machine Learning Engineer Intern

Taipei, Taiwan

- Trained VoiceCraft, a transformer based text-to-speech (TTS) model enabling voice synthesis and editing.
- Integrated customized phonemizer, VoiceCraft, and GAN-based vocoder to enhance voice quality.
- Developed a transformer based lyrics-to-melody model with custom data representation and embedding.
- Implemented countdown embeddings to align the number of melody notes with lyric words.

Institute of Information Science, Academia Sinica

Jul 2023 – Apr 2024

Research Assistant

Taipei, Taiwan

- Co-authored a paper on service demand extraction using large language models (LLMs), presented at SCC 2024.
- Increased input/output extraction accuracy from 56%/48% to 97%/98%, surpassing previous methods.

Projects

ToxicTone | Python, PyTorch

- Co-authored a paper titled ToxicTone as the first author, which was accepted for presentation at Interspeech 2025.
- Created the first large-scale Mandarin toxic speech dataset with prosodic labels.
- Multimodal models outperformed text-only baselines, highlighting the need for speech-based toxicity detection.

Bubblo | Unity, C#

- Led a team of four engineers and one artist to develop a 2D platformer with unique mechanics and original artwork.
- Designed and enforced a modular, low-coupling architecture using object-oriented programming principles.
- Adopted agile practices to prioritize tasks, meet deadlines, and iteratively improve gameplay and user experience.
- Check out the game here: [Bubblo](#)

Machine Problems | C, RISC-V

- Implemented user-level threading by allocating separate stacks in heap memory.
- Optimized virtual memory management through demand paging and swapping to maximize resource efficiency.
- Built xv6's timer interrupts to enable preemptive user-level threading and designed scheduling algorithms.
- Extended the inode structure to support double indirect blocks, increasing the maximum file size in the file system.

Technical Skills

Languages: C/C++, Python, JavaScript, RISC-V

Web Development: React.js, Node.js, Express.js, MongoDB, PHP, MySQL

DevOps: Docker, Kubernetes, Bash, Git, Conda, NVM

Machine Learning: Scikit-Learn, PyTorch, Wandb